Assignment 4: Building, and critically interpreting concept bottleneck models for skin lesion diagnosis prediction

Responsible AI 2026

In this project, you will build an intrinsically explainable concept bottleneck model (CBM) for diagnosing skin lesions (from photography) into five different cancer-related diagnostic categories. To this end, you will utilize the derm7pt dataset, which is annotated with seven check-list criteria (the 7-pt criteria) that you will use to form intermediate concepts for your model.

Upon building the CBM, you will establish a validation pipeline to assess the quality not just of the model, but also of its intrinsic explanations. Moreover, you will add further layers of interpretation and inspection to the model and extend your validation pipeline to also critically assess these.

Note: Parts 2-3 will be released in the upcoming weeks. Again, your project should be presented in the form of a poster.

Part 1: Building the CBM and creating explanation success criteria

- Download the derm7pt dataset and use an AI tool of your choice to **build a concept bottleneck model** that utilizes the 7-pt criteria as an intermediate concept layer and predicts the diagnostic labels. Make sure to verify that you can read the resulting code and that you find the resulting model to be in line with the definition of a concept bottleneck; you will be responsible for its correctness.
- Set your model up so that it can be **trained either with independent, sequential or joint training.** Again, make sure that this is done correctly, and ensure that you can account for the data allocated to training, validation and test, and that you agree with the choices.
- Train the model using all three regimes, and **perform a standard assessment of the models' performances**. What do you see?
- Perform an **exploratory study of the explanations given by the concepts**: Do they support any performance differences that you observe between the three training regimes? What happens if you intervene on the concepts as in the CBM paper?

- In discussion with your group members, suggest 3 potential use cases of model explanations for this type of prediction model. Summarize your discussion into a list of use cases and corresponding measurable success criteria.
    - **Example:** *Usecase* - Get patients to seek healthcare when needed. *Success criterion* - Patients using the AI tool are better at seeking healthcare when they need it, and staying away when they don't.